Room 0 — Caesar Cipher Warm-Up

Caesar Encryption with Known Key ($K = 4$)

Encrypting the original plaintext message with a Caesar cipher using a shift key of $K = 4$, while preserving whitespace and punctuation.

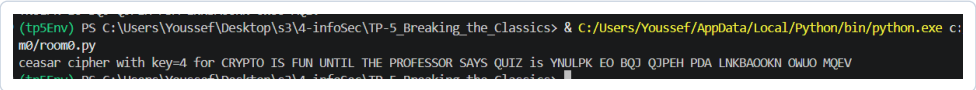


Figure 1: *Terminal output displaying the resulting ciphertext generated from the baseline plaintext with key* $K = 4$.

Caesar Decryption with Known Key ($K = 8$)

Decrypting the intercepted ciphertext using Caesar decryption with the known shift key $K = 8$ to recover the original message.

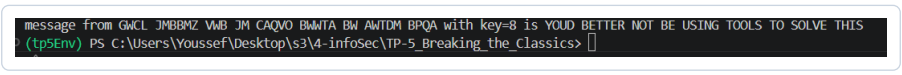


Figure 2: *Terminal output confirming successful decryption of the intercepted ciphertext using key* $K = 8$.

Exhaustive Brute-Force Decryption (Unknown Key)

When the secret key is unknown, an attacker can exploit the tiny keyspace of the Caesar cipher through an exhaustive brute-force attack. Testing all 26 candidate shifts generates a complete list of candidate plaintexts, from which the valid English sentence is readily identified by visual inspection.

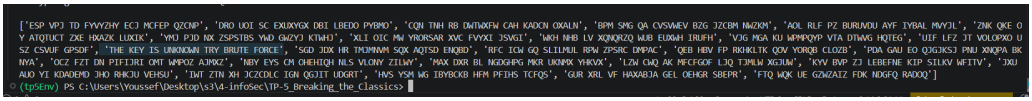


Figure 3: *Terminal output showing all* 26 *candidate plaintexts generated via brute force, highlighting the correct message at shift* 11.

Cryptanalytic Analysis: Caesar Cipher Vulnerability

The Caesar cipher provides zero practical confidentiality because its entire keyspace contains only $|\mathcal{K}| = 26$ possible shifts (effectively 25 non-trivial transformations). An attacker can compute and inspect every candidate plaintext in less than a millisecond without prior knowledge of the key. Because the shift is uniform across the entire message, the statistical frequency distribution of the underlying language is completely preserved, making automated cryptanalysis trivial even for longer texts.

Room 1 — Monoalphabetic Substitution Cipher

Substitution Encryption via Arbitrary Permutation Key

Encrypting the plaintext message using a fixed substitution alphabet, where each plaintext letter is mapped to a pre-assigned, pseudorandom ciphertext character.

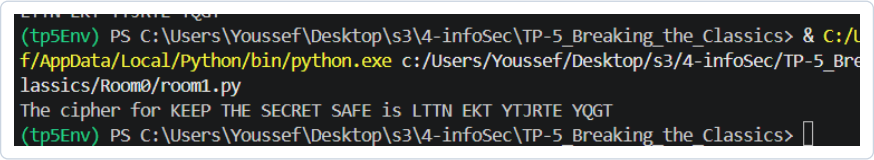


Figure 4: *Terminal output displaying the ciphertext produced using the defined 26-letter substitution mapping table.*

Substitution Decryption with Inverted Mapping

Decrypting an intercepted ciphertext by inverting the substitution mapping pairs (Cipher → Plain) and processing the characters through the decryption routine.

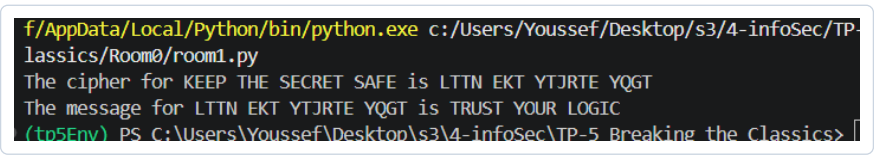


Figure 5: *Terminal output recovering the secret message (`"TRUST YOUR LOGIC"`) using the inverted substitution alphabet.*

Initial Automated Frequency Cryptanalysis

To attack an intercepted paragraph with an unknown key, an automated program computes the occurrence percentages of each ciphertext character and performs an initial rank-based alignment against standard English letter frequencies.

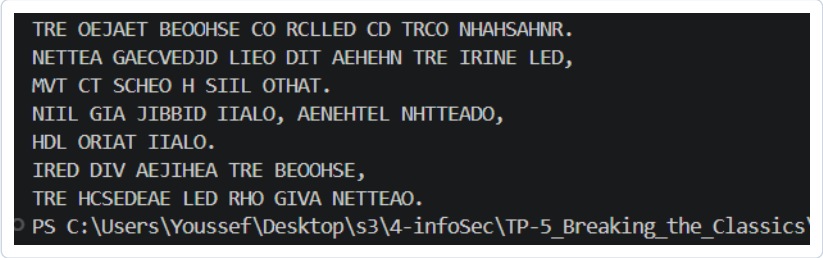


Figure 6: *Terminal output displaying the partially decrypted paragraph obtained from automated frequency ranking, providing an essential starting baseline.*

Hypothesis Formulation via Linguistic Patterns

Because sample frequencies in short paragraphs fluctuate, frequency analysis provides a strong hypothesis rather than the complete key. Combining the statistical ranking with recognizable short words (`EKT` → `THE`), repeated letters (`BTYYQFT` → `MESSAGE`), and common roots allows us to deduce the exact letter assignments.

```
assignedAlphabets2 = [
    ('A', 'Q'), ('B', 'M'), ('C', 'J'), ('D', 'Z'), ('E', 'T'),
    ('F', 'G'), ('G', 'F'), ('H', 'K'), ('I', 'P'), ('J', 'W'),
    ('K', 'L'), ('L', 'S'), ('M', 'B'), ('N', 'O'), ('O', 'X'),
    ('P', 'N'), ('Q', 'C'), ('R', 'R'), ('S', 'Y'), ('T', 'E'),
    ('U', 'V'), ('V', 'H'), ('W', 'I'), ('X', 'A'), ('Y', 'D'),
    ('Z', 'U')
]
```

Figure 7: *Terminal output showing the deduced list of letter mapping hypotheses based on frequency analysis and word structure.*

Full Plaintext Recovery & Clue Extraction

Applying the completed substitution mapping decrypts the entire intercepted text and reveals the essential clue needed to unlock the next room.

```
THE SECRET MESSAGE IS HIDDEN IN THIS PARAGRAPH.
LETTER FREQUENCY DOES NOT REVEAL THE WHOLE KEY,
BUT IT GIVES A GOOD START.
LOOK FOR COMMON WORDS, REPEATED PATTERNS,
AND SHORT WORDS.
WHEN YOU RECOVER THE MESSAGE,
THE VIGENERE KEY HAS FOUR LETTERS.
PS C:\Users\Youssef\Desktop\s3\4-infoSec\TP-5_Breaking_the_Classics\
```

Figure 8: *Terminal output displaying the fully recovered plaintext, confirming that the Vigenère key in Room 2 consists of four letters.*

Cryptanalytic Analysis: Substitution Ciphers & Frequency Analysis

Unlike the Caesar cipher, a general monoalphabetic substitution cipher features a massive keyspace of $26! \approx 4.03 \times 10^{26}$ possible permutations, rendering exhaustive brute-force attacks computationally impossible. However, the cipher suffers from a fundamental structural flaw: it is strictly monoalphabetic, meaning the mapping is one-to-one and deterministic. Consequently, the characteristic frequency profile of the underlying natural language (such as the high recurrence of `'E'`, `'T'`, and `'A'` in English) is directly transferred into the ciphertext. By combining unigram frequency rankings with contextual digraphs and repeated word patterns, the entire key space can be collapsed systematically.

Room 2 — Polyalphabetic Vigenère Cipher

Coset Decomposition & Frequency-Based Shift Recovery

Because the clue recovered from Room 1 revealed that the Vigenère key is exactly four letters long, characters located at indices sharing the same position modulo 4 ($i \equiv k \pmod 4$) are encrypted using identical Caesar shifts.

We partition the ciphertext into four distinct cosets (groups 1 to 4) and run frequency analysis on each:

- In English prose, the letter `'E'` is predominantly the most frequent character.
- By identifying the highest-frequency symbol in each group and measuring its modular distance to `'E'`, we determine the exact shift applied to that group.
- Decrypting each coset with its recovered shift and interleaving the characters back into their original positions fully reconstructs the plaintext.

Key Recovery, Full Decryption, & Access Word

Executing the automated coset recovery routine successfully recovers the four-letter keyword and restores the original message:

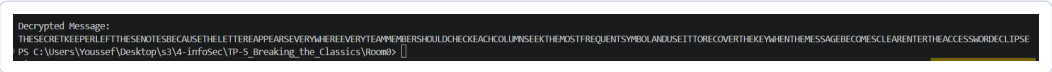


Figure 9: Terminal output logging the recovered four-letter Vigenère key (`"CODE"`) and the fully decrypted message, revealing the final access word `"ECLIPSE"`.

Cryptanalytic Analysis: Polyalphabetic Weakness & The Path to OTP

The Vigenère cipher was historically considered *le chiffre indéchiffrable* because alternating between multiple substitution alphabets flattens the single-letter frequency distribution. However, once the key length L is determined (whether via Kasiski examination, the Friedman Index of Coincidence, or side clues), the cipher completely collapses into L independent, parallel Caesar ciphers. Each coset can then be solved in isolation using standard unigram frequency analysis.

To achieve unconditional information-theoretic security, a cipher must ensure that the key is truly random, never reused, and as long as the message itself—principles that directly define the One-Time Pad (OTP).